

Phase I/II Operating Characteristics

New non-TITE alpha = 0 versus U-BOIN, BOIN12, and EffTox | updated one-stage rows

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No OBD %
Scenario 1						
DLT rate	0.3	0.35	0.4	0.45	0.5	
Efficacy rate	0.05	0.06	0.07	0.08	0.09	
OBD selection (%)						
New non-TITE one-stage	17	7.2	1.7	0.1	0.1	73.9
New non-TITE 2-stage highest utility	18	7.2	1.2	0	0.2	73.4
New non-TITE 2-stage top-2 randomized	17.4	7.2	2	0.8	0.1	72.5
U-BOIN	15.6	17.4	20.6	18.7	13.7	14
BOIN12	17.5	12	4.8	1.4	0.9	63.4
EffTox	5	4	6	3	6	75
Mean administrations by dose						
New non-TITE one-stage	11.1	4.4	2.3	0.9	0.3	
New non-TITE 2-stage highest utility	11.1	4.4	2.3	1	0.3	
New non-TITE 2-stage top-2 randomized	11	4.4	2.3	1	0.3	
U-BOIN	9	5.4	4.7	3.9	3.7	
BOIN12	10	5.8	3.3	1.9	0.8	
EffTox	8	3.2	3	2	2	
Scenario 16						
DLT rate	0.02	0.06	0.09	0.14	0.26	
Efficacy rate	0.2	0.2	0.2	0.2	0.2	
OBD selection (%)						
New non-TITE one-stage	29.3	21.3	20.2	18.8	10.4	0
New non-TITE 2-stage highest utility	34.1	21.3	16.6	16.4	11.5	0.1
New non-TITE 2-stage top-2 randomized	33.9	22.2	16.5	15.4	12	0
U-BOIN	28.1	26.8	21.1	16.3	7.7	0
BOIN12	35.1	27.5	19.7	12.1	5.6	0
EffTox	66	5	8	7	9	5
Mean administrations by dose						
New non-TITE one-stage	7.3	6.7	5.7	5.7	4.6	
New non-TITE 2-stage highest utility	7.6	6.8	5.7	5.5	4.4	
New non-TITE 2-stage top-2 randomized	6.6	6.1	5.6	6.1	5.6	
U-BOIN	6.1	6.2	6.4	6.4	5.4	
BOIN12	4.9	5.7	6.7	7.4	5.3	
EffTox	20.9	2.3	2.5	1.7	2.1	

Sources: updated New non-TITE one-stage alpha = 0, existing two-stage rows, U-BOIN, BOIN12, and EffTox N = 30 (1,000 simulations each).

No OBD % is source-specific: New non-TITE = No OBD; U-BOIN = Stop; BOIN12 = No selection; EffTox = None selected.

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Scenario 20						
DLT rate	0.07	0.14	0.31	0.41	0.52	
Efficacy rate	0.05	0.25	0.25	0.25	0.25	
OBD selection (%)						
New non-TITE one-stage	14.4	43.9	25.7	10.8	2.4	2.8
New non-TITE 2-stage highest utility	14.2	47.4	25.8	7.7	1.7	3.2
New non-TITE 2-stage top-2 randomized	10.4	47.7	25.6	10.6	2.7	3
U-BOIN	8.5	47.5	18.6	12.4	13	0
BOIN12	11.3	62.1	18.8	4.9	0.1	2.8
EffTox	24	30	20	7	10	9
Mean administrations by dose						
New non-TITE one-stage	5.2	11	7.6	4.2	1.4	
New non-TITE 2-stage highest utility	5.3	11.2	7.9	3.8	1.3	
New non-TITE 2-stage top-2 randomized	5.7	9.5	8	4.6	1.8	
U-BOIN	5.9	10.8	7	4	2.4	
BOIN12	6.2	10.7	7	4.1	1.7	
EffTox	13.5	6.6	5.1	2	1.7	
Scenario 24						
DLT rate	0.01	0.02	0.05	0.1	0.25	
Efficacy rate	0.05	0.08	0.2	0.5	0.5	
OBD selection (%)						
New non-TITE one-stage	1.4	0.9	5	58.6	34.1	0
New non-TITE 2-stage highest utility	2	1.5	6.2	55.5	34.8	0
New non-TITE 2-stage top-2 randomized	0.4	0.8	3.6	56.6	38.6	0
U-BOIN	1.5	3.2	8.2	60.1	27	0
BOIN12	1.7	3.3	6.7	65.7	22.6	0
EffTox	5	5	10	18	63	1
Mean administrations by dose						
New non-TITE one-stage	3.4	3.5	4.4	12.2	6.5	
New non-TITE 2-stage highest utility	3.4	3.5	4.4	10.6	8.1	
New non-TITE 2-stage top-2 randomized	3.6	3.7	4.9	9.4	8.4	
U-BOIN	3.5	4	5.1	9.8	7.9	
BOIN12	3.2	3.4	4.7	12.9	5.8	
EffTox	8.1	3.3	4.6	4	9.9	

Sources: updated New non-TITE one-stage alpha = 0, existing two-stage rows, U-BOIN, BOIN12, and EffTox N = 30 (1,000 simulations each).

No OBD % is source-specific: New non-TITE = No OBD; U-BOIN = Stop; BOIN12 = No selection; EffTox = None selected.

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	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No OBD %
Scenario 27						
DLT rate	0.11	0.17	0.3	0.5	0.62	
Efficacy rate	0.1	0.2	0.4	0.42	0.45	
OBD selection (%)						
New non-TITE one-stage	16.4	22.3	43.7	10.9	1.1	5.6
New non-TITE 2-stage highest utility	16.3	23.9	44	8.4	0.9	6.5
New non-TITE 2-stage top-2 randomized	10.5	23.6	46.6	12.5	1.1	5.7
U-BOIN	8.7	20.9	38.9	16.5	14.5	0.5
BOIN12	20.9	32.9	38	4.2	0	4
EffTox	36	27	15	12	6	4
Mean administrations by dose						
New non-TITE one-stage	7.1	7.9	9.8	3.6	0.7	
New non-TITE 2-stage highest utility	7.7	7.9	9.4	3.2	0.9	
New non-TITE 2-stage top-2 randomized	7.4	8.2	8.4	4	1.1	
U-BOIN	6.9	8.1	8.8	4.3	1.8	
BOIN12	7.2	9	9.2	3.4	0.7	
EffTox	16.6	5.1	3.7	2.2	1.7	
Scenario 38						
DLT rate	0.05	0.1	0.12	0.15	0.3	
Efficacy rate	0.05	0.08	0.12	0.15	0.5	
OBD selection (%)						
New non-TITE one-stage	14	7.9	6	7.8	63.2	1.1
New non-TITE 2-stage highest utility	17.9	10	7.3	7.5	56	1.3
New non-TITE 2-stage top-2 randomized	11.7	6	5.5	8.5	67.3	1
U-BOIN	6.2	8.9	13	13.3	58.6	0
BOIN12	13.2	13.8	16.5	13.6	41.5	1.4
EffTox	1	2	8	15	73	2
Mean administrations by dose						
New non-TITE one-stage	4.5	4.5	5	5.3	10.4	
New non-TITE 2-stage highest utility	4.8	4.7	5.3	5.4	9.6	
New non-TITE 2-stage top-2 randomized	4.7	4.8	5.3	6.9	8.2	
U-BOIN	5.2	5.5	5.6	6.3	7.8	
BOIN12	4.2	4.7	5.2	6.1	9.6	
EffTox	3.4	3.3	4.5	5.2	13.3	

Sources: updated New non-TITE one-stage alpha = 0, existing two-stage rows, U-BOIN, BOIN12, and EffTox N = 30 (1,000 simulations each).

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Mean Unique Patients by Method

Mean number of unique patients per simulated trial

	Scenario 1	16	20	24	27	38	Mean
New non-TITE one-stage	12.3	18.2	18.9	19.6	19.4	18.9	17.9
New non-TITE 2-stage highest utility	12.4	18.2	18.9	19.6	19.4	18.8	17.9
New non-TITE 2-stage top-2 randomized	12.4	18.3	19	19.5	19.5	18.5	17.9
U-BOIN	26.7	30.5	30.1	30.3	29.9	30.4	29.7
BOIN12	21.8	30	29.7	30	29.5	29.8	28.5
EffTox	18.2	29.5	28.9	29.9	29.3	29.7	27.6